

Activity pattern analysis by means of sequence-alignment methods

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Abstract. The author describes a method of comparing sequences of characters, called sequence alignment or string matching, and illustrates its use in the analysis of daily activity patterns derived from time-use diaries. It allows definition of measures of similarity or distance between complete sequences, called global alignment, or the evaluation of the best fit of short sequences within long sequences, called local alignment. Alignments may be done pairwise to develop similarity or distance matrices that describe the relatedness of individuals in the set of sequences being examined. Pairwise alignment methods may be extended to many individuals by using multiple alignment analysis. A number of elementary hand-worked examples are provided. The basic concepts are discussed in terms of the problems of time-use research and the method is illustrated by examining diary data from a survey conducted in Reading, England. The CLUSTAL software used for the alignments was written for molecular biological research. The method offers a powerful technique for analyzing the full richness of diary data without discarding the details of episode ordering, duration, or transition. It is also possible to extend the analysis to include the context of activities, such as the presence of other persons or the location, but such extensions would require software designed for social science rather than biochemical problems. The method also offers a challenge to researchers to begin to develop theories about the determinants of daily behavior as a whole, rather than about participation in single activities or about time-budget totals.

1 Introduction

How much information is left on the shelf when activity research is based on time budgets and not on detailed episodic data provided by time-use diaries? The detailed perspective retains all information provided by survey respondents to diary surveys. The more common time-budget studies generalize the details of activity diary episodes into 24-hour days containing total times devoted to a set of activities. The problem of information waste was addressed by Rydenstam (1994) in his paper on event history analysis.

Studies of activity episodes may be described as micro and qualitative, because they preserve the detail offered by survey data, tend to concentrate on smaller numbers of people, and use literal rather than quantitative analytical methods. The research objectives in the detailed perspective tend to be focused on small groups, and often relate to the factors determining the sequence of activities, their location, transitions from one to another, and even the subject's experience of stress or satisfaction.

Research employing the detailed perspective implicitly deals with activity sequences. At any time scale, from the single day, to the life cycle, to multigenerational family history, human life can be described as a sequence of acts that Hägerstrand (1970; 1975) illustrated as paths in a space with dimensions for time, geographical space, and activity. According to this model, individuals' activity patterns are the outcome of their pursuit of personal objectives or projects within an environment of constraints and opportunities created by other individuals and by social and economic institutions. A considerable body of research into activity patterns (often called the Lund School) emanated from this time-geographic perspective during the 1970s and 1980s, some of which was collected by Carlstein et al (1978).

Time-budget studies may be described as both macro and statistical because they concentrate on aggregate data of populations. The research objective is often to describe and explain the differences in time-allocation patterns among subpopulations. Details of the order, transitions, episode duration, and contexts of activities are ignored. Diary surveys have been employed for most of this century to generate time-budget data. Time budgets have been used to provide general sociological descriptions (Sorokin and Berger, 1931; Meier, 1959; Krantz, 1970), to examine particular social problems such as the relationship between time allocation and parental stress (Sanik, 1993), and to derive land-use planning implications from ordinary human behaviour (Chapin, 1974; Michelson, 1975; 1993). Harvey (1982) took the time-space constraint system as his point of departure in examining the social role and activity context of discretionary time use. He incorporated the location, duration, presence of other persons, and time of day as critical aspects in his analysis, but based his research on time budgets, acknowledging that activity sequence was an omitted key aspect. Recently an issue of the journal *Social Indicators Research* (Harvey, 1993) was devoted to time-use research. A comprehensive, but now slightly dated, review of time-use research is that by Parkes and Thrift (1980). Recently, many research projects have been directed at the measurement of household production. This work has led to the development of satellite accounts as supplements to national income accounting programs to estimate the relative sizes of the formal and informal economies (Ironmonger, 1989; Pyatt, 1990; Swinamer, 1985).

Sequence analysis has been used in many disciplines to study behaviour in human and animal societies. The elements of human or animal activity sequences are often categorical states such as life-cycle stages, marital status, courting behaviour, etc. Quantitative methods have been extended to research on sequences since the 1950s (Gottman and Roy, 1990), but Elliot (1978) observed that there was still a scarcity of methodological literature that applied to daily activity sequences. He reviewed applications in biology and other natural and social sciences. Abbott (1995) categorized analytical methods in sequence analysis as applying to sequence steps or to whole sequences. Step-by-step methods examine relationships among elements or states and use information-theoretic, autoregressive, or transition probability methods. Whole sequence methods use the entire sequence as the unit of analysis and concentrate on questions of global similarities often with a view to establishing sequence typologies. In an intermediate situation, local similarities are analyzed to assess the role of key subsequences embedded in longer strings of elements. This is frequently the case in genetics, where comparatively short subsequences of proteins have crucial informational or controlling roles. Few applications of whole sequence analysis exist in social science. The earliest that I know of are those by Abbott and Forrest (1986), Abbott and Hrycak (1990), and more recently Halpin and Chan (1995). There appear to be no applications of local sequence analysis in the social sciences.

Whole sequence methods using what are called alignments were developed in computer science (text editing and voice recognition) and molecular biology (protein and DNA analysis). Sankoff and Kruskal (1983) wrote the basic text on the subject. Because the data gathered for time use and behavioural research are inherently sequential, the methods of sequence-alignment analysis would seem to be appropriate for activity pattern studies. The use of such methods offers the possibility of using well-defined concepts of similarity and difference on the full richness and complexity of activity diaries. Applications or alignment methods in molecular biology use comparatively simple alphabets (the four nucleotide molecules, or the twenty amino acids) but may have very long sequences, in the order of thousands. Applications of sequence analysis in time-use studies may use shorter sequences but the alphabet representing activities could have dozens or even hundreds of letters.

When processes or events are recorded as sequences we may ask questions such as whether two are the same or not, how similar they are, or whether parts are similar. Sequential data may be inherently discrete, or may be sampled at intervals from continuous processes. Elliot discussed the practical difficulties of defining activity episodes when the definition of activity includes contextual data. For the purpose of this paper, activity episodes are simply those provided by survey respondents in their diaries, but some suggestions for including context are made later.

The general procedure used in sequent alignment is to define a number of elementary operations on the alphabet for amending the sequence. The simplest case analyzes two sequences, and operations are repeated until the two sequences are made equivalent. When few operations are necessary, it means that two sequences are similar. More operations mean that the sequences are less similar. The elementary operations are insertion–deletion pairs, and substitutions. Some advanced research involves more complex operations.

2 Vocabulary of alignment analysis

2.1 Pairwise alignment

Pairwise alignments involve two sequences, called the source and the target. The target is written below the source so that equivalent elements, called matches, are in alignment. Null elements, sometimes called gaps, are inserted into each to equalize their lengths and to move equivalent elements into alignment. Alignments can be evaluated in terms of similarity or distance measures. Alignment analysis is carried out by an algorithm that either maximizes similarity or minimizes distance. There are generally many ways to align sequences, some being better than others in terms of similarity or distance measures, and frequently optimal alignments are not unique. When multiple optima exist, the selection of particular optima as the basis for further analysis or theorizing is a matter of judgment, not method.

2.2 Elementary operations

The most elementary operations are insertions and deletions. The sequence [ad] may be converted into [add] by inserting a 'd' at the end or by deleting the last 'd' from [add]. Either way, the alignment is illustrated as follows:

```
ad-
add
```

The insertion of a character in one sequence is equivalent to deleting it from the other, meaning that the operations are always used in pairs. In fact the operation is called an indel by some authors. The aligned sequence, called the consensus, is different from the source and target because it is a hybrid or combination of the two.

Substitutions may also be allowed or may be achieved by a pair of insertions and deletions. For example, [at] may be converted into [it] by the substitution of 'i' for 'a' or by inserting 'i' after 'a' as follows:

```
a-t
-it
```

We evaluate the quality of the alignment by assigning weights or scores to the operations. The relative weights assigned to substitutions and insertion–deletion pairs depend on the particular research topic. The researcher may construct the weights to reflect the interpretability of either operation as part of the social process under examination.

Certain advanced applications define compression and expansion of sequences (other than using insertion–deletion pairs) and transpositions. An acceptable analysis consists of operations using the defined set. An analysis may apply constraints to the use of operations, such as limiting the number of adjacent insertions or substitutions.

2.3 Similarity, distance, and weight

Algorithms for finding optimal alignments have used both similarity and distance measures to define the optimum (Needleman and Wunsch, 1970; Sellars, 1974). Smith et al (1981) have shown that in certain circumstances the two approaches are equivalent. When the analysis is based on distances, every elementary operation has a weight greater than or equal to zero, with matches scoring 0. If the analysis is based on similarities, matches receive higher scores. Weights may be tailored to reflect the importance of operations, the similarity of particular elements, the position of elements in the sequence, or the number or type of neighboring elements or gaps. Distance measures may be defined so as to satisfy the metric axioms (Sankoff and Kruskal, 1983, page 22) and consequently inferences that use such properties may be made concerning sequences. An example of the use of such a property is the statement that the distance from sequence *a* to *c* is less than or equal to the sum of the distances from *a* to *b* plus *b* to *c*. The concepts of weight and similarity (or distance) are related because a weight for an operation is a similarity (or distance) between two sequences of length 1.

2.4 Globally optimal alignment

When whole sequences of roughly similar length are analyzed, the alignment is called global. The dynamic programming algorithms referenced above were devised independently during the early 1970s in biology and computer science to find the minimum number of operations to align sequences. Biological research using the ideas of similarity of sequences had been common since at least the 1960s, but the algorithm of Needleman and Wunsch (1970) was the first to give an explicit means of evaluating sequence similarity. This finding was the key to determining which alignments are optimal and to counting the number of optimal alignments. Sankoff and Kruskal (1983), States and Boguski (1994) and Waterman (1995), give full accounts of the basic algorithm. This section is based on States and Boguski, but for programmable algorithms see Waterman.

The globally optimum alignment of two sequences which have a consensus of length *n* is the score of the optimum for an alignment of length (*n* – 1) plus the score for the alignment of the final element. This can be expressed as a recursion relation in which the similarity score for any optimum alignment is defined by the previous optimum plus the current operation. As an example consider the alignment of [waiter] and [water]. The definition of the optimal alignment requires the comparison of every pair of elements. To do this the sequences are arranged in a matrix as in figure 1. The simplest similarity measure scores a 1 for matching elements and 0 for mismatches or gaps. Beginning at the top-left cell, the cell scores are defined as the sum of the score for the element pair, defined by the row and column heading, plus the sum for the best alignment preceding the pair being evaluated. This will be found somewhere in the row or column band above and to the left of the cell being evaluated. The first row and column have no preceding alignments so the scores for these cells are 1 for matches and 0 for mismatches.

The score for cell (2,2) is the sum of the score for the match of [a] in water with [a] in waiter (in this case 1) plus the score for the previous alignment (also 1), making 2. The remaining cells in the [a] column and the [a] row score 0 for mismatches plus 1 for the previous match of [w]. The best previous score is always the largest score in the row or column above and to the left of the cell being evaluated.

	w	a	i	t	e	r
w	1	0	0	0	0	0
a	0	2	1	1	1	1
t	0	1	2			
e	0	1				
r	0	1				

Figure 1. Partial scoring table for the alignment of [waiter] and [water].

	w	a	i	t	e	r
w	1*	0	0	0	0	0
a	0	2*	1	1	1	1
t	0	1	2	3*	2	2
e	0	1	2	2	3*	2
r	0	1	2	2	2	4*

Figure 2. Completed scoring table for the alignment of [waiter] with [water].

The score for cell (3,3) is thus 2, comprised of 0 for the mismatch plus 2 for the matches previously achieved. Figure 2 gives the completed matrix.

The actual alignment is constructed by following a path from cell to cell through the table by a process called a traceback. The path begins at the highest scoring element in the final row or column. This may or may not be at the corner, and the first step moves back to the highest scoring cell in the previous row or column. The path continues through the highest scoring elements of previous rows or columns to the origin, or to an element in the first row or column. The steps of the path are indicated in figure 2 by asterisks. When the path skips one or more rows to join a cell higher up in the previous column, gaps of varying lengths are created. The best previous score is often not unique, so the best path frequently splits into fairly complex networks. Gaps correspond to insertion–deletion operations discussed earlier. A path may join matched, mismatched, or gapped elements. The optimal alignment is as follows:

waiter
wa-ter

An example from the Reading survey is shown in figure 3. The diaries of a 68-year old woman, nicknamed Mrs Yeats, were available for seven days. The scoring matrix

Thursday (21 activities)	Wednesday (20 activities)																			
	r	c	e	d	e	t	s	t	c	e	l	p	p	r	c	e	q	v	c	r
r	1*	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1
p	0	1*	1	1	1	1	1	1	1	1	1	2	2	1	1	1	1	1	1	1
e	0	1	2*	1	2	1	1	1	1	2	1	1	2	2	2	3	2	2	2	2
d	0	1	1	3*	2	2	2	2	2	2	2	2	2	2	2	3	3	3	3	3
d	0	1	1	3	3	3*	3	3	3	3	3	3	3	3	3	3	3	3	3	3
s	0	1*	1	2	3	3	4*	3	3	3	3	3	3	3	3	3	3	3	3	3
e	0	1	2*	2	4	3	3	4*	4	5	4	4	4	4	4	4	4	4	4	4
d	0	1	1	3*	3	4	4	4	4	4	5	5	5	5	5	5	5	5	5	5
i	0	1	1	2	3	4	4	4	4	4	5	5	5	5	5	5	5	5	5	5
c	0	2	1	2	3	4	4	4	5*	4	5	5	5	5	6	5	5	5	6	5
e	0	1	3	2	4*	4	4	4	4	6*	5	5	5	5	5	7	6	6	6	6
c	0	2	2	3	3	4	4	4	5	5	6	6	6	6	7*	6	7	7	8	7
g	0	1	2	3	3	4*	4	4	4	5	6	6	6	6	6	7	7	7	7	8
s	0	1	2	3	3	4	5*	4	4	5	6	6	6	6	6	7	7	7	7	8
g	0	1	2	3	3	4	4	5*	5	5	6	6	6	6	6	7	7	7	7	8
c	0	2	2	3	3	4	4	5	6*	5	6	6	6	6	7	7	7	7	8	8
e	0	1	3	3	4	4	4	5	5	7*	6	6	6	6	6	8*	7	7	7	8
c	0	2	2	3	3	4	4	5	6	6	7	7	7	7	8*	7	8	8	9	8
q	0	1	2	3	3	4	4	5	5	6	7	7	7	7	7	8	8	8	9*	8
v	0	1	2	3	3	4	4	5	5	6	7	7	7	7	7	8	8	10*	9	9
r	1	1	2	3	3	4	4	5	5	6	7	7	7	8	7	8	8	9	10	11*

Figure 3. Wednesday–Thursday activity alignment of Mrs Yeats.

and alignment are shown in figure 3 for her Wednesday and Thursday diaries. An account of the survey and the key to the activity codes in the row and column headings are given in section 5.

The Wednesday and Thursday activity sequences have a similarity score of 11. Because the maximum number of possible matches is related to the sequence length, scores are often expressed as percentages of the average sequence length. In this case the score converts to 11/20.5 or about 54% similarity.

Two optimal paths are illustrated in the table, indicated by highlighted elements on the path. All of the alternative paths between the two shown are also optimal paths, indicating that there are very many optima. The two paths, and hence alignments, shown are of particular interest. One is formed by joining the highest or right-hand eligible cell at each step of the traceback. The other is formed by joining the lowest or left-hand cell. These are called the highroad and lowroad paths, respectively, and they bound the space in the matrix containing all optimal alignments. The highroad and lowroad are:

Highroad alignment:

Wednesday rcedetst--celpprc----e-qvcr

Thursday rped-dsedice----cgsgcecqv-r

Lowroad alignment:

Wednesday r----ced--e-tstcelpprceqvcr

Thursday rpeddshedicecgsgce----c-qv-r

Other optimum alignments begin and end like the highroad and lowroad but are achieved by using different patterns of gaps in the central portions of the sequences. The Wednesday and Thursday activity patterns are moderately divergent from morning to the evening meal, which tends to spread the highroad and lowroad away from the principal diagonal and creates the numerous optimal alignments. Certain other days in Mrs Yeats's diary have longer similar sequences at the beginning and end, which narrows the variable period and reduces the number of optima.

Readers will have realized that random sequences may generate nonzero similarity scores and that it is essential to have some evidence that optimal alignments produced by the algorithm are better than random alignments. We want to know the expected score for alignments of two randomly generated sequences of elements having the distribution of Mrs Yeats's activities over the two days shown in figure 3. Given the simple 0–1 scoring system and the additional restriction on the weighting function that no gaps are allowed, the expected score is estimated by the summation of the squared relative frequencies for all activities multiplied by the sequence length. This assumes that each successive activity is independent of previous activities. The summation of the squared relative frequencies is 0.12 (see appendix) which is the expected match score for 2 activity sequences of length 1. For a sequence of 21 activities, the expected number of matches is 2.5.

It is clear that the optimal alignment score is far larger than the expected score of randomly matched activities. However, the distributions of similarity or distance scores are not known for most sequences and weighting functions so it is not possible to determine the exact significance of differences between optimal alignment scores and scores of random sequences. Researchers have often compared optimal alignment scores with scores from Monte Carlo simulations of random sequences with the same distribution of elements as appear in the real sequences to estimate the significance of such differences. The original paper by Needleman and Wunsch (1970) included such a

simulation. A description of some research into the distribution theory of sequence scores is given by States and Boguski (1994).

2.5 Dot matrix methods

Globally optimal alignments may be inappropriate where sequences are of different lengths or radically different form. In such cases, dot matrix methods or local alignment algorithms are available. Where short fragments are to be compared to longer sequences, the fragment may be split into unpredictable subsequences and scattered along the longer sequence rather than being aligned with nearly exact parts. This occurs in the alignment of [waiter] with [water everywhere] which has three optima. One produces:

```
wait-----er-----
wa-tereverywhere
```

The other two simply move the [er] group ahead to match the first [er] group in the long sequence, or back to match the last. It is possible to design gap penalties to adjust for this, but these may affect some other part of complex alignments.

The other situation where global alignment may not be appropriate is where extensive subsequences are identical but are rearranged. The alignment of [wherever] with [everywhere] produces:

```
-----wherever
everywhere---
```

This is optimal in the sense of the definitions underlying the algorithms, but a method that emphasizes the reversal of (wher) and (ever) is desirable. The dot matrix method is older than alignment algorithms and is often used in conjunction with alignments because the strengths and weaknesses of each are complementary.

The matrix is formed by writing the two sequences across the top and down the side of a table like the one used for the global alignment. Dots are placed in cells where letters match. A self-comparison of a single sequence produces a square table with a continuous pattern of dots on the main diagonal. Random matches will be scattered throughout the table when letters occur more than once. When sequences are of different lengths but contain similar subsequences or are rearrangements of the same phrases, the diagonals are displaced in a rectangular table, but strong diagonal patterns are preserved. The [wherever] [everywhere] alignment is shown in figure 4.

The similarity of the two sequences is illustrated by the negatively sloped diagonal patterns in the upper-right and lower-left areas of the matrix. The inversion of the fragments [where] and [ever] in the sequences is indicated by the displacement of the diagonals. It will be noted that both diagonals are intersected by positively sloped diagonal patterns. These arise because the two sequence fragments being aligned

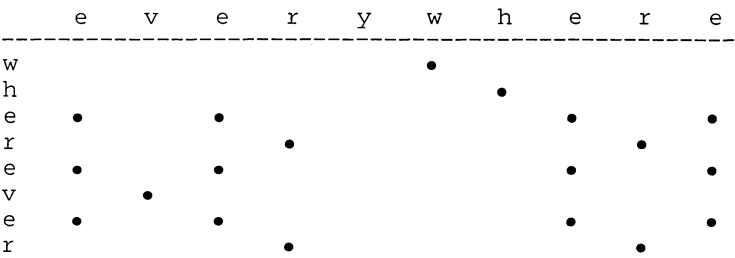


Figure 4. Dot matrix alignment of reversed sequences.

contain portions [eve] and [ere] that are the same when arranged backwards or forwards. Such palindromes are interesting because they show that activity elements making up certain projects are perfectly symmetrical.

Dot matrix methods have been extended to accentuate the graphic patterns that they produce and to define quantitative scores for sequence similarity. However, the method is fundamentally graphical and its use with the moderately large sample sizes available in diary surveys would require one to examine very large numbers of pairwise comparisons. An extensive account of dot matrix methods and local and global algorithms is presented by States and Boguski (1994). They recommend that dot matrix methods be used as a starting point for many types of analysis in molecular biology.

2.6 Local alignment method

Smith and Waterman (1981) developed an amendment to the global dynamic programming algorithm that stops the search for additional elements in an alignment by giving negative scores to mismatches and gaps. The highest scoring path does not have to begin in the first row or column and continue to the last row or column. This means that the optimal alignment is not required to use all elements of both sequences and it concentrates on finding highly similar alignments of subsequences. This solves the first of the problems with global methods described above but does not address the reversal problem. Still, the use of local alignments may be useful in finding similar parts of daily activity patterns that are common among people even when the overall activity patterns are dissimilar. Figure 5 shows the scoring matrix for the search for the local alignment of the sequence pair: [thereinatimefor], [astitchintimesaves].

The local optimum alignment is:

inetime
in-time

Sequence 2	Sequence 1														
	t	h	e	r	e	i	n	a	t	i	m	e	f	o	r
a	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0
s	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0
t	10	0	0	0	0	0	0	0	10	0	0	0	0	0	0
i	0	5	0	0	0	10	0	0	0	20	5	0	0	0	0
t	10	0	0	0	0	0	5	0	10	5	15	0	0	0	0
c	0	5	0	0	0	0	0	0	0	5	0	10	0	0	0
h	0	10	0	0	0	0	0	0	0	0	0	0	5	0	0
i	0	0	5	0	0	10*	0	0	0	10	0	0	0	0	0
n	0	0	0	0	0	0	20*	5	0	0	5	0	0	0	0
t	10	0	0	0	0	0	5	15	15*	0	0	0	0	0	0
i	0	5	0	0	0	10	0	0	10	25*	10	5	0	0	0
m	0	0	0	0	0	0	5	0	0	10	35*	20	15	10	0
e	0	0	10	0	10	0	0	0	0	5	20	45*	30	25	20
s	0	0	0	5	0	5	0	0	0	0	15	30	40	25	20
a	0	0	0	0	0	0	0	10	0	0	10	25	25	35	20
v	0	0	0	0	0	0	0	0	5	0	0	20	20	20	20
e	0	0	10	0	10	0	0	0	0	0	0	10	15	15	15
s	0	0	0	5	0	5	0	0	0	0	0	0	5	10	10
n	0	0	0	0	0	0	15	0	0	0	0	0	0	0	5
i	0	0	0	0	0	10	0	10	0	10	0	0	0	0	0
n	0	0	0	0	0	0	20	5	5	0	5	0	0	0	0
e	0	0	10	0	10	0	5	15	0	0	0	15	0	0	0

Scores: match = 10; mismatch = -5; gap (length *k*) = -(10 + 5*k*).

Figure 5. Local alignment calculation matrix.

This alignment is found by locating the maximum element in figure 5, which is 45 at the match of 'e' in cell (13, 12), and tracing its path to its origin. If the first sequence were [thereisatimefor], the optimal subsequence is simply [time]. The misprint of [in] for [is] illustrates the situation where an initial match score is reduced by a later mismatch then increases by additional matches. Once a long sequence of mismatches is encountered, the score declines. The optimal subsequence is then indicated by the path from the matrix cell with the maximum score back to (but not including) 0. This defines the element pairs of the local optimum alignment.

2.7 Multiple alignments

When two sequences are involved, the time required by algorithms is proportional to the square of the average sequence length. It is often useful to examine more than two sequences and this is called multiple alignment. This would probably be the usual situation in time-use studies. Needleman and Wunsch (1970) discussed the extension of the pairwise alignment to multiple sequence problems but they did not discuss the resource requirements. The time needed for multiple global alignment algorithms is in the order of the n th power of the average sequence length, where n sequences are analyzed. Such multiple alignments become prohibitively demanding on computer resources for only moderately large problems. Waterman (1989) has shown that the number of different alignments of only a few biological sequences (where the number of nucleotides or amino acids exceeds 1000) easily exceeds the number of elementary particles in the universe. A large number of sequences in biology is generally much smaller than the numbers of diary days typically recorded in even moderately large time-use surveys.

Much research has involved finding assumptions that allow algorithms to use fewer steps to find optimal alignments (Carillo and Lipman, 1988; Gotoh, 1982), and developing algorithms that find near-optimal alignments in very few steps (Lipman and Pearson, 1985). However, the most commonly used strategy for obtaining multiple global alignments is to develop a matrix of pairwise similarities or distances, and then to apply some form of clustering procedure. In biology, the clustering is hypothesized to be a model of the evolutionary history of the biochemistry of the tissue being analyzed. It is not clear at present how one would interpret the clustering strategy in a social science application.

As with the global case, the natural extension of research using pairwise local alignments is to find multiple local alignments. For an account of one approach see Schuler et al (1991).

2.8 Discrete sequences sampled from continuous processes

Sometimes the process of interest is continuous but the analyst has decided to extract a sample of the characteristic at discrete time intervals. This situation, called time warping, arises in speech pattern recognition and in acoustics. The principles of sequence analysis of time warps are discussed by Kruskal and Liberman (1983).

The basic difference time warping and alignment is that the time warp involves expansion of elements from a shorter sequence portion into a longer portion, or compression from the longer into the shorter. Gaps are not allowed or needed as the shorter sequence fills the longer one. Time-warped elements may map to more than one element in the other sequence (expansion) or may have multiple elements mapped from the other sequence (compression). The weights employed in such many-to-one mappings must be related, whereas the weights assigned to aligned elements are independent of penalties assigned to gaps that lead or follow them.

It may be that time warping is a more appropriate method for analyzing episodic diary data because it does not require the introduction of gaps, which are not part of the raw data, nor an interpretation of their meaning. Time warping may also accommodate

the variation in the level of detail provided by different survey respondents more effectively than alignments. These are topics for further research and are not examined in this paper.

3 Sequence analysis of activity patterns

In this section I develop some similarities between sequence-analysis problems in molecular biology and in time-use studies as a prelude to application of such methods to activity pattern research. The basic concepts and vocabulary of sequence alignments are used to develop applications to activity research. Examples of comparisons of daily diaries are presented in section 5.

3.1 Elemental activities

The Reading diary survey asked respondents what they were doing and when. This generated a large number of literal descriptions of activities which were preserved in as much detail as seemed reasonable. For example, 'drinking coffee' was coded separately from 'having tea', but was coded the same as 'having coffee'. The fact that the respondent offers the beginning time and a new description of an activity was considered *prima facie* evidence that the action was sufficiently significant to warrant a separate code and record.

Some surveys use precoded data-capture forms on which interviewers enter their view of the closest match to the respondent's description. Here the investigator determines the system of elemental activities and their categorization at the survey planning stage. Many such surveys ask for contextual data for each diary entry, such as who else was present, where did it happen, did you feel stress, and so forth. Such data add immensely to the usefulness of diary data, particularly in interpreting the personal or social significance of particular actions. However, such information complicates the problem of activity definition. For example, does the activity change if a companion leaves?

The simplest solution is to include contextual data explicitly in the definition of elemental activities as additional dimensions of the data space. Such dimensions could contain a number of categories. For example, the presence of other persons could be classed as: alone, family, or other persons with or without family. Each of n activities would then be coded into one of the three categories, expanding the activity classification to $3n$ classes. As more contextual dimensions are added, the coding system expands as the product of the number of categories in each dimension. This solution may be adequate for initial investigative applications, but is very demanding on the alphabet used to represent elementary activities. In addition, many of the possibilities in the potential data matrix will be empty.

3.2 Tree matching

The situation discussed above is really a result of the fact that time-use data augmented by contextual data has a tree structure, not simply a string or sequence structure. The tree may be considered to have the activity sequence as its central trunk, but the activity elements or nodes in fact have at least twigs (if not branches) attached containing various details of situational information. This suggests that tree matching or editing techniques may be more appropriate in the long term. Tree editing proceeds by defining the same elementary operations as are used in string editing (substitutions, insertions, and deletions). These are applied to nodes and leaves of one tree successively until it is transformed into a second tree. As with sequence matching, the distance between trees is defined as the minimum number of edit operations necessary to transform one into the other.

Noetzel and Selkow (1983) explain tree editing, but I do not pursue this extension in this paper. In the short term, contextual data may be incorporated by explicitly

containing them in the elemental activity. For example, watching television alone could be defined as a different activity than watching television with others. The Reading survey contained detailed locational data but did not ask for details about other persons or personal feelings so the limitation is not a problem.

3.3 Length

Typical diaries contain from 10 to 40 separate activity episodes, where an episode is a period of activity from a beginning to an ending time. The episode is the coarsest recording interval for activities. It is significant in the sense that respondents themselves identify episodes as the building blocks of daily activity where data are gathered in a free format diary. Given the beginning and ending times of an episode, it is possible to define sequences of the same elementary activity each lasting any length of time and filling the episode duration. The 1-minute interval is probably the shortest practicable sequence element and generates a very fine activity record, 1440 elements per day, while remaining manageable. This may be an appropriate choice because it weights activities by their duration. If the recording grid were coarsened to, say, 5 or 10 minutes, the situation falls somewhere between the polar cases.

The length of sequences defined as reported activity episodes is variable but quite short (usually less than 40), which makes minimal demands on computer memory. With such short sequences, analysis would be comparatively quick. The example in figure 3 was calculated by hand then checked with a spreadsheet that took less than two hours to design and program.

The use of 1-minute intervals for recording activities increases the computer demands greatly and introduces problems of representing sequences and alignments. The aligned sequences are inherently graphic and 1440 characters times what may be a large number of diary days produces an awkward volume of material to be illustrated as an alignment. Intervals of 5 to 15 minutes are more manageable.

3.4 Global and local analysis

The applications to time-use research that first come to mind are taxonomic. Definition of similarities or distances between individual activity patterns allows the use of statistical procedures such as cluster analysis for definition of similar groups and classifications. One might ask global questions about whole daily activity patterns such as:

How similar are the activity patterns of men and women?

Are the patterns of the unemployed more like the retired than the employed?

Is my Tuesday activity pattern closer to Wednesday than to Sunday?

Have the activity patterns of women changed significantly over the past ten years?

Projects, as defined by the Lund School, consist of activity sequences designed to achieve well-defined tasks. An example is the rising – washing – dressing – eating sequence common in the morning. Projects may comprise slightly different sequences of similar activities and last for varying lengths of time. The analysis of projects is more suited to local than to global alignments. One may ask:

Is my routine for leaving the house in the morning like yours?

Are shopping trips mainly the same?

Is a trip to the opera similar to a trip to a baseball game?

4 Software

Simple pairwise alignments can be carried out by hand or on spreadsheets, but when we want to align several sequences, computer software is necessary. Time-use studies in the social sciences typically survey several hundreds or thousands of respondents. Although it may not be necessary to examine whole data sets simultaneously, it seems clear that multiple alignment analyses of time-use data will require at least as much

computational power as applications in the natural sciences. Software packages developed by molecular biologists are abundant but there are few for social science applications. As the saying goes, "Neither borrower nor lender be; but if you have to be one, be a borrower". Borrowing from research in molecular biology means that we must make do with alphabets of 4 letters (the nucleotides) or 20 letters (the amino acids). This will not be satisfactory in the longer term, but 20 activities are sufficient to illustrate some applications to activity pattern and to begin to develop hypotheses about activity sequences suitable for testing using alignment methods.

Multiple alignment methods are less than about 10 years old in molecular biology. The majority of software packages reviewed in recent survey articles such as those by McClure et al (1994) or Chan et al (1992) have been published since 1987. The examples presented in this paper use the CLUSTAL programs of Higgins, reported in Higgins and Sharp (1988), and in Thompson et al (1994).⁽¹⁾

CLUSTAL performed satisfactorily in the evaluations cited above in comparison with other packages. It runs on Unix, Vax, Mac, and Dos operating systems, and allows the user to select from an extensive range of input parameters to tailor the analysis as needed. Among these are separate penalties for introducing and extending gaps in the sequences, and for weighting mismatches (penalizing substitutions). The latter feature is particularly useful because it allows the user to specify a weight matrix and thus to avoid using the specialized default-weighting systems based explicitly on biochemical criteria. These are obviously desirable for the intended applications but have no application in time-use studies. Specialized weighting schemes will have to be developed in time-use research to allow substitutions of relatively similar activities (say reading for stamp collecting) while penalizing different activities (say automobile racing for child care).

CLUSTALV, used in this paper, reads a file of activity sequences and produces two files. One file specifies a dendrogram that shows the similarity of sequences as they are joined into a consensus, and the other is the final alignment of the sequences. A newer, more flexible, and powerful version, CLUSTALW documented by Thompson et al (1994), has replaced the dendrogram with a file defining an evolutionary tree.

The dendrogram file is not graphic. It contains the similarity measures that define the joining pattern of the individual sequences into the final combined alignment. A multiple alignment usually requires one step less than the number of sequences being aligned. The dendrogram therefore defines the steps by which successive sequences are added to the overall alignment. At each step a sequence is aligned either to another sequence or to a consensus created by previous alignments of sequences and the similarities are recalculated. At the completion, the consensus is defined which contains all of the gaps and mismatches that were necessary to achieve the optimum alignment of all input sequences.

The alignment file consists of the characters representing the activities of the sequences. It shows the pattern of matches, mismatches, and gaps (created by insertions and deletions) needed to achieve the final consensus.

5 Examples from the 1973 Reading Urban Study

The Martin Centre for Urban and Architectural Research at Cambridge University undertook the Reading Diary Survey with funding from the Economic and Social Research Council (ESRC). Individuals in households selected through a multistage cluster sampling procedure were asked to complete diaries for a week. Over 2600 diary days were supplied by 450 respondents aged from 15 to 75. The database is available as item 1503 in the ESRC archive at the University of Essex.

⁽¹⁾ The software is available from the European Molecular Biology Laboratory through the internet (www.embl-heidelberg.de). Select computational services, biological information services, then databases of molecular biology software.

The survey is documented by Bullock et al (1974). It was undertaken as part of a program of research projects called the Reading Urban Study that was active at the Martin Centre during the early 1970s. The research focused on investigating the relationships between land use, transportation systems, and activity patterns (see *Architectural Design* 1971; Martin and March, 1972; Shapcott and Steadman, 1978; Tomlinson et al, 1973).

To illustrate the methods of sequence comparisons in activity pattern studies, I will use seven diaries of a respondent to whom I shall refer by the pseudonym, Mrs Yeats. The Reading Urban Study did not survey families so we have only the diaries of individuals in sampled households. Mrs Yeats and her husband own their own home, but do not own a car. Mr Yeats is retired. Daily diary activities were aggregated into nineteen general categories for analysis (see table 1).

Table 1. Classification of nineteen activities.

sleep and rest (r)	eating and drinking (e)	full-time education (f)
private leisure (l)	watching television (q)	sport (a)
travel, public modes (g)	community service, religious	organized leisure,
casual social (v)	observation, organizations (n)	entertainment (m)
domestic work (d)	employment, main or second	child, family care (k)
personal care, hygiene (p)	job (w)	cooking and washing
information, reading,	travel, private modes (t)	dishes (c)
study (i)	shopping, use of services (s)	unknown activity (y)

Mrs Yeats's daily activities expressed as character sequences are as follows:

Wednesday rcedetst0celpprceqvcr

Thursday rpeddsed0icecgsgecqvcr

Friday rpeciddptnt0ceciststdlcecqldepr

Saturday rpecddp0cecilvcecyptvmtpr

Sunday rpecddptn0tcrceicicectntqecqr

Monday rcecdtstdc0eecidlcecqtitqeqr

Tuesday rcecddd0ctitcecpgvtpqtitveqr

A sleep episode has been added to the beginning of each day's diary as respondents were instructed to begin each diary with the first activity after waking. The 0 in the string indicates noon. The 0 is not recognized by CLUSTAL and does not enter into any of the analysis that follows. Table 2 shows the pairwise percentage similarity scores for the daily activities for Mrs Yeats.

Table 2. Percentage similarity matrix for Mrs Yeats's seven diary days (short form).

	Wed	Thur	Fri	Sat	Sun	Mon
Thur	30					
Fri	25	28.6				
Sat	25	38.1	25			
Sun	35	42.9	37.0	45.8		
Mon	25	42.9	39.3	37.5	33.3	
Tues	30	28.6	39.3	33.3	33.3	57.1

Note: Diagonal elements are 100%.

5.1 One person, multiday, short-form activity sequences

The multiple alignment shown in figure 6 is based on the pairwise percentage similarity scores. The alignment shows the respondent number, the day, and the number of activities recorded in the diary as a diary day label. The right-hand part of the figure shows the activity alignment. Mrs Yeats's diaries consist of five recognizable activity sequences or projects: the morning meal (eating and washing up), morning housework, morning errands, noon meal, and the evening meal. These are indicated by the stars and dots in the last row of figure 6. These projects were identified by inspection, not by calculation, and their locations should be considered to be approximate.

911 wed	19	rce---detstcelpprce-----qvcr
911 thur	20	rpe-dd-sedicecgsgce-cqv-----r
911 fri	28	rpeciddptntcecitstdlceqcldopr
911 sat	23	rpecdd-p---cecilvce-cyptvmtpr
911 sun	26	rpecdd-ptntcrceceice-ctntqecqr
911 mon	27	rcecdtstdceecidlce-cqtitqeqr
911 tues	27	rcecdddctitcecpgvtv-pqtitveqr
		... ***... *

Figure 6. CLUSTALV multiple sequence alignment for Mrs Yeats's seven-day diary. (Gap penalties: open = 10, extension = 10).

Mrs Yeats's first activity every day is either washing and dressing or making breakfast, followed by eating and washing up. Five of seven days begin with the [pec] or [cec] sequence. This is followed by up to three episodes of housework ending on five days with an episode of personal care or having tea. One day she made a purchase from a mobile grocery van before having the tea break. Morning work is thus regularly punctuated by some sort of break.

On five days a trip follows the morning housework sequence, sometimes a shopping trip, but on Friday, she went to church. On Tuesday she made a short trip to use the pay telephone, as she does not have a telephone in her house. The travel episodes are not aligned as regularly as the meal and housework sequences. The alignment identifies the noon meal sequence [cec] very clearly. Afternoons are a variable period with fewer matches identified.

The final period of common activity is the evening meal, which is perfectly aligned on four days and is partially present on Wednesday when washing up is postponed by other activities. The evenings are variable. Mrs Yeats sometimes goes out visiting, makes telephone calls on two days, goes to church on Sunday evening (in addition to Sunday morning), nearly always watches television, and frequently prepares and eats snacks. However, these do not align regularly for more than three days.

The dendrogram files, which define the steps by which daily activity sequences are aligned, are shown in the appendix. The files show daily sequences or groups of sequences that are aligned at each step and the percentage similarity of the sequences in the consensus at that step. The file consists of a matrix of zeros, ones, and twos. Days indicated by 0 do not align at a given step. When two days align, they are designated as 1 and 2. If a consensus joins another consensus constructed in a previous step, all days comprising each consensus are designated as 1 or 2.

Mrs Yeats's alignment begins with the highest pairwise score in table 2 which is Monday with Tuesday. At the next step, the weekend days join each other. At the third step, Thursday joins the weekend group, then Friday joins the weekday group. The weekday and weekend groups join at the fifth step, leaving Wednesday to join all other

days at step 6. Wednesday seems to be unlike all other days by the absence of morning washing up after breakfast, only one morning domestic work episode, and the presence of afternoon personal care and rest episodes.

5.2 One person, multiday, long-form activity sequences

Table 3 gives the percentage similarity scores for the same diary sequences (that is to say, the lower triangle of table 2), called the short form, and the diaries coded into 5-minute sequences, called the long form. The long input sequences are all 288 characters initially, but the consensus alignment is 300 characters, accommodating all gaps. The product-moment correlation coefficient of the long-form and short-form similarity scores is 0.02. Clearly, the decision to examine episodes independently of duration (that is, the decision to use short sequences) is of fundamental importance and generates different similarity patterns among days.

Table 3. Short-form and long-form percentage similarity matrix for Mrs Yeats's seven-day diary.

	Wed	Thur	Fri	Sat	Sun	Mon	Tues	
Wed		64.2	50	48.3	51.7	48.3	64.6	} long form
Thurs	30		55.9	49.3	51.7	56.6	68.1	
Fri	25	28.6		55.9	59.7	60.4	54.2	
Sat	25	38.1	25		48.3	51.0	56.6	
Sun	35	42.9	37.0	45.8		54.9	52.1	
Mon	25	42.9	39.3	37.5	33.3		60.1	
Tues	30	28.6	39.3	33.3	33.3	57.1		
short form								

Correlation coefficient (short, long) = 0.02.

Figure 7 (see over) shows that, although the similarity matrix has changed substantially and the alignment is now over ten times longer, Mrs Yeats's days still contain identifiable projects: morning meal, housework, noon meal, evening meal, early evening television, visiting, and late evening television. The activity period based on trips in the morning has been absorbed into housework because of the short duration of travel and out-of-home activity, telephone calls in particular.

The weighting of activities by their duration introduces variability in the starting times of days, which upsets the simple alignment of the morning projects that were apparent in the short-form sequences. However, even the variable times and durations of rising, dressing, cooking, and eating do not erase the regularity of the daily breakfast period.

The morning housework period is even more clearly demarcated by the introduction of activity durations because the activity is both long and relatively regular in its starting time. The end of the housework period is more ragged because of the variability of the housework durations and of the punctuating personal care and eating activities. The shopping and church travel remain nearly in alignment, but the walk to the telephone box disappears completely because it is only 2 minutes long. The trip on Monday occurs too early to remain in alignment with the Wednesday, Friday, and Sunday trips.

The noon meal period is also shown more clearly in the long-form sequences because the meal preparation time is illustrated. The noon meal project now occupies a major portion of the alignment at midday.

Afternoon newspaper reading and the crossword also produce a common activity period in the early afternoon because of their length. Omission of duration hides this similarity in the short-form sequences. The evening meal sequence is shorter than the

Figure 7. CLUSTALV multiple alignment for Mrs Yeats's long-form diaries (gap penalties: opening = 10, extension = 10).

noon meal sequence and is pushed somewhat out of alignment by long afternoon activities, but it remains visible to some extent.

Three common evening periods have appeared again because of the long durations of evening activities, especially television and visiting. Television watching tends to occur after supper, and again later in the evening after visiting and just before going to bed.

The joining steps for the long-form diaries begin with Thursday and Tuesday, the largest element in the upper triangle of table 3. This group is joined by Wednesday at step 2. At step 3, Friday and Monday from a second weekday group, which is joined by Sunday at step 4. The connection of Sunday with the Friday group arises in part by church attendance in the morning of both days. Step 5 joins the two groups created so far, and step 6 adds Saturday to complete the clustering. The pattern is different from the joining of the short-form diaries in that the weekend days do not form a clear grouping. The most similar days also change from Monday–Tuesday in the short-form diaries to Thursday–Tuesday. Evidently, the analysis is dependent on the initial decision to use short-form or long-form diaries. It will be important to establish the conditions under which each of the two data formats is appropriate.

5.3 Four-person, multiday, short-form activity sequences

This example is based on the alignment of the weekday diaries of four widowed women aged from 67 to 73. All had incomes in the two lowest income groups on a five level scale. Together they provided 18 diary days. Their diaries are designated by their respondent identification numbers (225, 803, 2515, and 3111) and weekday. Weekend diaries were omitted from this example because the method seems very sensitive to weekday/weekend differences based on the analysis of Mrs Yeats’s diaries. We want to see how well alignment methods discriminate among diaries without the comparatively easy identification of weekend days.

As the number of diary days increases, it becomes more difficult to analyze the alignment (shown in figure 8) and the joining pattern of the dendrogram file becomes the natural point of departure.

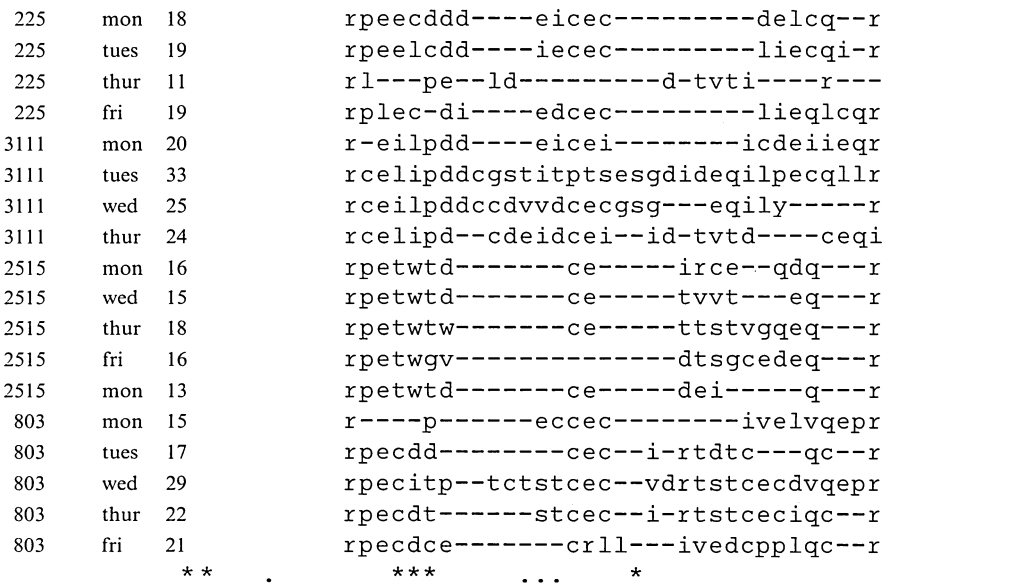


Figure 8. CLUSTALV multiple alignment for four persons, 18 diary days (gap penalties: opening = 10, extension = 10).

The steps of the clustering process are as follows:

1. 2515 first Monday joins with second Monday.
2. 2515 Wednesday joins with Friday.
3. 803 Tuesday joins with Thursday.
4. 225 Monday joins with Tuesday.
5. 2515 Monday group (step 1) joins with Wednesday group (step 2).
6. 803 Wednesday joins with Tuesday group (step 3).
7. 2515 Friday joins with all other days (step 5).
8. 225 Thursday joins with 3111 Thursday.
9. 225 Friday joins with Monday group (step 4).
10. 3111 Tuesday joins with Wednesday.
11. 3111 Monday joins with 803 Monday.
12. 225/3111 Thursday group (step 8) joins with 803 group (step 6).
13. 225 groups (step 9) joins with 3111 group (step 10).
14. 3111/803 Monday group (step 11) joins with 225/3111 group (step 13).
15. 3111/8033/225/3111 group (step 14) joins with 225/3111/803 group (step 12).
16. 2515 group (step 7) joins with 803 Friday.
17. All diaries—step 15 with step 16.

Seven most-similar groups are created for individual respondents before any diary day is joined between different people. It is therefore clear that the method finds a great number of personal similarities before it joins activity patterns across individuals. In fact after the joining of the Thursday diaries of persons 225 and 3111, two further connections of individual diaries occur before a further mixed person consensus appears. The pattern is apparent in the consensus alignment shown in figure 8 which illustrates the similarity of days for individuals.

Respondent 2515 has a job as a cleaner and the extended duration of job episodes creates long gaps in the middle of her activity sequences (see figure 8). She undertakes fewer activities than the other three women, averaging just under 16 per day. Her mornings are virtually identical, consisting of: washing and dressing, eating breakfast, travel to work, work, travel home, domestic chores, and cooking and eating the midday meal. The afternoons show some variability with travel and shopping or visiting on Wednesday and Thursday. All days end with television before she goes to bed.

Respondent 225 averages nearly 17 activities per weekday. Her morning rising sequence is not precisely defined, but begins either with personal care or care for her pet, which is coded as a leisure activity. She then eats breakfast and washes up. The morning continues with housework episodes and is punctuated by either eating or reading a newspaper (coded as information). The noon meal sequence aligns on three of four days. Her afternoons and evenings are weakly structured. The afternoon is variously composed of housework, pet care, visiting, reading, and walking. Thursday, she spent the whole day from mid-morning to 6 PM out visiting. The uniqueness of Thursday is demonstrated by the dendrogram which joins Monday, Tuesday, and Friday, but adds Thursday to one of the days of respondent 3111.

Respondent 3111 engages in the largest number of activities, averaging over 25 a day. Her morning meal sequence is regular and is followed every day by a leisure–information or information–leisure sequence composed of reading and listening to the radio. This is followed by washing and dressing and housework. The remainder of her mornings are complex, containing up to 5 activities before the noon meal sequence. The evening meal is reported every day but it does not align with itself. The evenings regularly contain at least one episode of television viewing, usually immediately after the meal. The rest of the evening is occupied by a range of activities including leisure (usually music), domestic work, personal care, and eating. The activity pattern of

the weekdays of respondent 3111 is considerably more complex and diffuse than the patterns of 2515.

Respondent 803 is also fairly active, averaging just under 21 activities per day. Mornings consist of washing and dressing, eating, washing dishes, and trips or housework. The noon meal sequence is present every day. She follows the noon meal with either letter writing (information) or visiting, after which she often rests. The rest period is followed by a trip to the post office to post letters or by leisure, information, or visiting. Television, snacks, and personal care comprise evening activities. Monday breakfast was misaligned with the noon meal sequences of the other days. This is probably a consequence of the use of exploratory gap opening and extension penalties. More sophisticated penalties based on tests of the software with additional example diaries will probably improve the alignments.

Although figure 8 is dominated by individual similarities, a few periods of activity are common to over half of the diaries. These are morning personal care and meal, morning housework, the noon meal, an afternoon trip, and evening television. The sample diaries exhibit common features both within and across individuals. It is probable, though, that the commonalities across individuals will be harder to identify as larger samples are examined.

6 Further research

The representation of daily activities as character sequences is an obvious technique for representing data, but the application of alignment algorithms to sequence analysis adds immense analytical potential to time-use studies. Abbot (1995) and Gottman and Roy (1990) foresee sequence analysis becoming perhaps the major method of examining social phenomena that occur over periods of time. This certainly includes time-use studies. There is a huge literature in molecular biology that will certainly assist in the application of alignment methods to social research problems.

A number of problems and questions lie ahead and have to be dealt with before alignment methods can reach their potential. These will include:

- (1) developing criteria for determining whether short-form or long-form episode representations are appropriate;
- (2) developing weighting schemes that describe the significance of activity substitutions;
- (3) including wider contextual data in activities, perhaps through tree matching methods;
- (4) developing tests for significance of similarity scores;
- (5) developing hypotheses about the factors that generate activity patterns as opposed to hypotheses about participation in individual activities;
- (6) developing software that supplies an alphabet sufficiently rich to represent social conditions in general, and daily activities in particular;
- (7) deciding whether to concentrate on similarity analysis of dendrograms or on the visual and sequential data provided by the actual alignments; and
- (8) defining research problems suited to local and to global alignment methods.

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References

- Abbott A, 1995, "Sequence analysis: new methods for old ideas" *Annual Review of Sociology* **21** 93–113
- Abbott A, Forrest J, 1986, "Optimal matching methods for historical sequences" *Journal of Interdisciplinary History* **16** 471–494
- Abbott A, Hrycak A, 1990, "Measuring resemblances in sequence data: an optimal matching analysis of musicians careers" *American Journal of Sociology* **96** 144–185

- Architectural Design* 1971, May, pp 275 – 322
- Bullock N, Dickens P, Shapcott M, Steadman P, 1974, "Time budgets and models of daily activity patterns" *Social Trends* number 5, pp 1 – 18
- Carillo H, Lipman D, 1988, "The multiple sequence alignment problem in biology" *Journal of Applied Mathematics (SIAM)* **48** 1073 – 1082
- Carlstein T, Parkes D, Thrift N (Eds), 1978 *Human Activity and Time Geography* (Edward Arnold, London)
- Chan S, Wong A, Chiu D, 1992, "Comparative analysis of multiple protein sequence alignment methods" *Bulletin of Mathematical Biology* **54** 563 – 598
- Chapin F S, 1974 *Human Activity Patterns in the City: Things People Do in Time and Space* (John Wiley, New York)
- Elliot D, 1978, "Constraints and sequences: elements of formal models of activity patterns", paper presented to Ad Hoc Group 10, "Time budgets and social activities", Ninth World Congress of Sociology, Upsala, Sweden; copy available from elliott@ac.dal.ca
- Gotoh O, 1982, "An improved algorithm for matching biological sequences" *Journal of Molecular Biology* **162** 705 – 708
- Gottman J, Roy A, 1990 *Sequential Analysis: A Guide for Behavioural Researchers* (Cambridge University Press, Cambridge)
- Hägerstrand T, 1970, "What about people in regional science?" *Papers and Proceedings of the Regional Science Association* **24** 7 – 21
- Hägerstrand T, 1975, "Survival and arena" *The Monadnock* **49** 2 – 29; published in *Human Activity and Time Geography* 1978, Eds T Carlstein, D Parkes, N Thrift (Edward Arnold, London) pp 122 – 145
- Halpin B, Chan W, 1995, "Optimal matching of moderately many class careers", WP 96-5, ESRC Research Centre on Micro-social Change in Britain, Essex University, Colchester, Essex
- Harvey A S, 1982 *Studies in Broadcasting 18. Role and Context: Shapers of Behaviour* (Nippon Hoso Kyokai, Tokyo)
- Harvey A S (Ed.), 1993 *Social Indicators Research* **30** (2 and 3)
- Higgins D, Sharp P, 1988, "CLUSTAL: a package for performing multiple sequence alignment on a microcomputer" *Gene* **73** 237 – 244
- Ironmonger D, 1989 *Households Work* (Allen and Unwin, Sydney)
- Krantz P, 1970, "What do people do all day?" *Behavioural Science* **15** 286 – 291
- Kruskal J, Liberman M, 1983, "The symmetric time-warping problem: from continuous to discrete", in *Time Warps, String Edits, and Macromolecules: The Theory and Practice of Sequence Comparison* Eds D Sankoff, J Kruskal (Addison-Wesley, Reading, MA) pp 125 – 161
- Lipman D, Pearson W, 1985, "Rapid and sensitive protein similarity searches" *Science* **227** 1435 – 1441
- McClure M, Vasi T, Fitch W, 1994, "Comparative analysis of multiple protein sequence alignment methods" *Molecular Biology and Evolution* **11** 571 – 592
- Martin L, March L, 1972 *Urban Space and Structure* (Cambridge University Press, Cambridge)
- Meier R, 1959, "Human time allocation: a basis for social accounts" *Journal of the American Institute of Planners* **25** 27 – 33
- Michelson W (Ed.), 1975 *Behavioural Research Methods in Environmental Design* (Dowden Hutchinson and Ross, Stroudsburg, PA)
- Michelson W, 1993, "Grounding time-use in microspace: empirical results" *Social Indicators Research* **30** 121 – 137
- Needleman S, Wunsch C, 1970, "A general method applicable to the search for similarities in the amino acid sequence of two proteins" *Journal of Molecular Biology* **48** 443 – 453
- Noetzel A, Selkow S, 1983, "An analysis of the general tree-editing problem", in *Time Warps, String Edits, and Macromolecules: The Theory and Practice of Sequence Comparison* Eds D Sankoff, J Kruskal (Addison-Wesley, Reading, MA) pp 237 – 252
- Parkes D, Thrift N, 1980 *Times, Spaces, and Places: A Chrono-geographic Perspective* (John Wiley, Chichester, Sussex)
- Pyatt G, 1990, "Accounting for time use" *Review of Income and Wealth* **36** 33 – 52
- Rydenstam K, 1994, "Who takes care of the household work after work?", paper presented at the 16th meeting of the International Association for Time Use Research, Bielefeld; copy available from k.rydenstam@scb.se
- Sanik M, 1993, "The effects of time allocation on parental stress" *Social Indicators Research* **30** 175 – 184
- Sankoff D, Kruskal J (Eds), 1983 *Time Warps, String Edits, and Macromolecules: The Theory and Practice of Sequence Comparison* (Addison-Wesley, Reading, MA)

Schuler G, Altshcul S, Lipman D, 1991, "A workbench for multiple alignment construction and analysis" *Proteins: Structure, Function, and Genetics* **9** 180 – 190

Sellers P, 1974, "An algorithm for the distance between two finite sequences" *Journal of Combinatorial Theory* **16** 253 – 258

Shapcott M, Steadman P, 1978, "Rhythms of urban activity", in *Human Activity and Time Geography* Eds T Carlstein, D Parkes, N Thrift (Edward Arnold, London)

Sorokin P, Berger C, 1931 *Time Budgets of Human Behaviour* (Harvard University Press, Cambridge)

Smith T, Waterman M, 1981, "Identification of common Molecular subsequences" *Journal of Molecular Biology* **147** 195 – 197

Smith T, Waterman M, Fitch W, 1981, "Comparative biosequence metrics" *Journal of Molecular Evolution* **18** 38 – 46

States D, Boguski M, 1994, "Similarity and homology", in *Sequence Analysis Primer* Eds M Gribskov, J Devereux (Oxford University Press, New York) pp 89 – 157

Swinamer J, 1985, "The value of household work in Canada 1981" *Canadian Statistical Review* March vi – xiv

Thompson J, Higgins D, Gibson T, 1994, "CLUSTALW: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position specific gap penalties, and weight matrix choice" *Nucleic Acids Research* **22** 4673 – 4680

Tomlinson J, Bullock N, Steadman P, Taylor E, 1973, "A model of students' daily activity patterns" *Environment and Planning A* **5** 231 – 266

Waterman M (Ed.), 1989 *Mathematical Methods for DNA Sequences* (CRC Press, Boca Raton, FL)

Waterman M, 1995 *Introduction to Computational Biology* (Chapman and Hall, London)

APPENDIX

Table A1. Relative frequencies of Mrs Yeats's 41 activities on Wednesday and Thursday.

Activity ^a	Relative frequency	Squared relative frequency
Sleep	0.12	0.014
Cooking	0.20	0.040
Eating	0.20	0.040
Domestic	0.10	0.001
Public travel	0.05	0.003
Private travel	0.05	0.003
Visiting	0.05	0.003
Information	0.02	0.000
Television	0.05	0.003
Leisure	0.02	0.000
Personal	0.07	0.005
Shopping	0.07	0.005
Total	1.0	0.12

^a See table 1.

Table A2. Dendrogram files for examples in section 5 of Mrs Yeats's diaries.

Step number	Percentage similarity	Day ^a	Step number	Percentage similarity	Day ^a
		wt f s s my			wt f s s my
(a) Short form			(b) Long form		
1	57.1	0000012	1	68.1	0100002
2	45.8	1200002	2	64.4	1200002
3	40.5	0102200	3	60.4	0010020
4	39.3	0010022	4	57.3	0010210
5	33.3	0121122	5	53.4	1120221
6	28.3	1222222	6	51.6	1112111

^a Tuesday is indicated by letter y.

Table A3. Dendrogram files for examples in section 5 of eighteen short-form diaries of four women.

Step number	Percentage similarity	Respondent and day ^a			
		225	3111	2515	803
		mytff	mywt	mwtfm	mywtff
1	64.3	0000	0000	10002	00000
2	62.5	0000	0000	01200	00000
3	61.1	0000	0000	00000	01020
4	57.9	1200	0000	00000	00000
5	57.7	0000	0000	12201	00000
6	46.1	0000	0000	00000	01210
7	43.6	0000	0000	11121	00000
8	41.7	0010	0002	00000	00000
9	41.1	1102	0000	00000	00000
10	38.5	0000	0120	00000	00000
11	37.5	0000	1000	00000	20000
12	33.4	0010	0001	00000	02220
13	32.2	1101	0220	00000	00000
14	29.5	1101	2110	00000	20000
15	28.7	1121	1112	00000	12220
16	27.0	0000	0000	11111	00002
17	26.4	1111	1111	22222	11112

^a Tuesday is indicated by letter y.